FlakeXplain

Evidence, pilot results and research direction

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Development assessment - not an untouched final evaluation. The original pilot and the follow-up use the same frozen projects and cases. No result below establishes superiority over CANNIER.

Projects	Model-ready cases	Observed positives	Positive prevalence
35	96,118	693	0.7210%

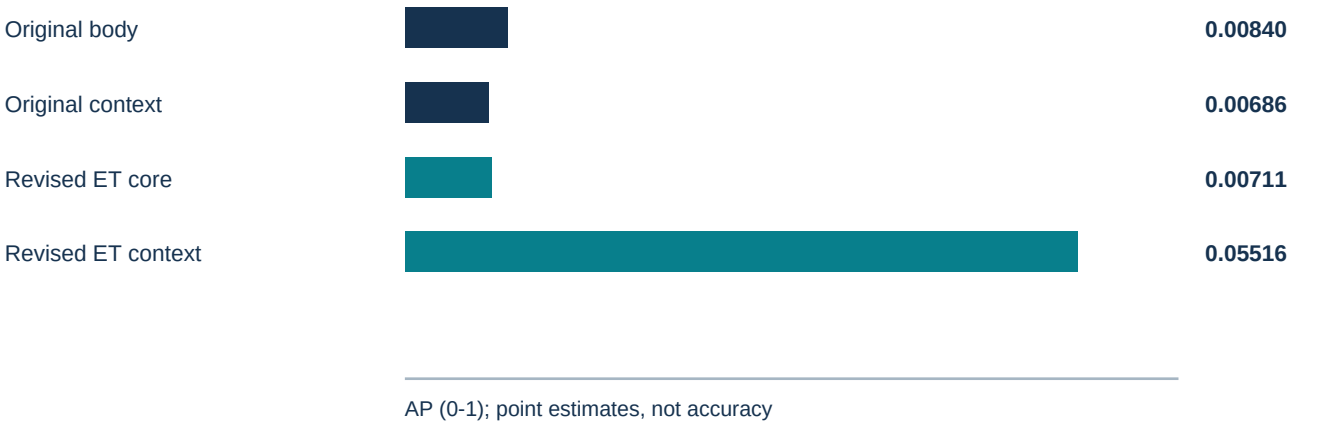
What has been completed

An evidence audit and exact-source joins; a frozen 22/55-feature pilot; a 100-case error-audit sample with 12 inspected source excerpts; a revised 30/96-feature representation; ten follow-up model variants with project-separated inner selection; saved predictions, models and verification records.

Measured outcome

The highest observed development AP is 0.05516 (Tuned ET: context v2), versus 0.00840 for the original body-only pilot. The predeclared revised-context comparison changes pooled AP by +0.04804, with a conditional project-bootstrap interval [-0.00651, +0.08545]. These are development findings on already-inspected folds, not proof of improvement over a published base paper.

Average precision comparison



At a top-10% budget separately within each project, captured positives are **63 original body / 103 revised core / 121 revised context**, out of 693 (9,628 selected cases). This is retrospective label capture, not measured rerun savings.

The highest AP is identified after development inspection. Pooled scores mix fold-model calibration. Conditional paired uncertainty is reported on page 2.

Controlled model comparison

All rows use 96,118 cases. P/R/F1 use thresholds selected only from inner training-fold predictions. AP is threshold-independent.

Model	AP (0-1)	Precision	Recall	F1	MCC	FP / 1k
Original ET: body	0.00840	1.24%	29.44%	2.37%	0.0278	170.7
Original ET: context	0.00686	0.84%	60.32%	1.66%	0.0149	515.3
Tuned ET: body v1	0.00615	0.53%	4.62%	0.95%	-0.0058	62.7
Tuned ET: context v1	0.03155	1.55%	47.62%	3.00%	0.0521	220.1
Tuned ET: core v2*	0.00711	1.51%	10.68%	2.64%	0.0215	50.7
Tuned ET: context v2	0.05516	1.33%	7.22%	2.25%	0.0146	38.8
LR: body v1	0.00987	0.41%	8.95%	0.79%	-0.0157	156.9
LR: context v1	0.02654	0.49%	14.29%	0.94%	-0.0143	212.1
LR: core v2*	0.01262	0.97%	8.66%	1.75%	0.0078	64.1
LR: context v2	0.01328	0.79%	22.37%	1.53%	0.0041	204.1
Text LR: body	0.00953	0.90%	22.94%	1.73%	0.0099	183.9
Text LR: context	0.00863	1.02%	21.79%	1.96%	0.0152	152.9

V2 core* includes six new body-operation counts and two scoped-read indicators in addition to the original 22 body counts. Thus v2 is an incremental-context comparison, not a pure body/context ablation. The v1 comparisons retain the strict 22-versus-55 distinction.

Revised Extra Trees comparison by outer development fold

Fold	Core v2 AP	Full context v2 AP
0	0.05792	0.11149
1	0.00927	0.00691
2	0.00435	0.01235
3	0.00421	0.07427
4	0.00654	0.00732

Primary context-minus-core AP: **+0.04804**. Conditional 95% project-bootstrap interval: [-0.00651, +0.08545] (500 resamples; no refitting). Project AP improved in 16 of 29 positive projects and regressed in 13.

What the error audit teaches us

A declared sample of 50 false positives and 50 false negatives spans 35 projects. Twelve source excerpts from 12 projects were checked. These are AI-assisted static observations; no causal annotations or label changes were made.

Example	Observed issue / representation consequence
Airflow False positive	Mock decorators supply arguments that appear as unresolved fixture candidates. Unittest assertions require call-aware counts.
Dask False positive	Fixed datetime.date values can trigger a time-API proxy despite deterministic intent. A risk token is not a failure cause.
Filelock False negative	Process events, timed waits and joins motivate blocking/cleanup features; runtime scheduling remains unobserved.
Freezegun False negative	Late imports and sys.modules checks expose module-cache state beyond the function body.
Hydra False negative	Parameterized cases share a body; parameter-specific behavior is not fully represented.
Xonsh False negative	Shell fixture context reaches the original extraction bound; dynamic behavior remains partial.

Exact numeric feature collisions

Representation	Features	Positive cases sharing vectors with negatives
v1_body	22	379 / 693 (54.7%)
v1_context	55	132 / 693 (19.0%)
v2_body	30	277 / 693 (40.0%)
v2_context	96	123 / 693 (17.7%)

Full-corpus labels are used only for this descriptive audit. Opposite labels with identical vectors cannot be perfectly separated by that representation. Fewer collisions demonstrate richer source information, not proven generalization.

Verification

466 checks passed, with 50 selected models replayed. Saved inner predictions reproduce configuration selection and thresholds. Training-only scaler/IDF fits and independent sklearn metrics were verified. The repository suite passed 84 tests; one Linux-only check was skipped. Convergence warnings: 0.

Research direction and decisions

Recommended scope

Fixture-aware static prediction of observed Python test instability across projects. Treat the current work as a reproducible development study. Retain negative results and report project/family sensitivity alongside pooled scores.

Research role	Reference and fit
Primary framing	A Large-Scale Empirical Investigation Into Cross-Project Flaky Test Prediction (IEEE Access, 2024): directly addresses project heterogeneity and adaptation.
Methodological baseline	Static Test Flakiness Prediction: How Far Can We Go? (EMSE, 2022): static metrics, studied in Java; requires a declared Python adaptation.
Python/hybrid reference	CANNIER (EMSE, 2023): runtime features and ML-assisted rerunning with different targets/budgets; not reproduced by this static experiment.

The 2024 adaptation settings and target-data assumptions must be audited before comparison. Its published scores cannot be compared directly with ours. A changed base-paper framing does not establish novelty or an experimental win.

Decisions for the guide

1. Agree the static cross-project task and primary reference.
2. Select a compatible released-baseline adaptation and a bounded fixture/parameter audit.
3. Freeze the final evaluation plan. Fresh independent evidence is needed for an untouched confirmation claim after these inspected folds.
4. Decide whether the main contribution is predictive improvement, an empirical replication/benchmark, or a validated analysis of representation limits.

Claims we can make now

Audited empirical corpus; reproducible negative reference and development comparisons; source-linked error examples; measured representation collisions. We cannot yet claim a superior CANNIER detector, confirmed OD causes, a complete fixture resolver, independent final performance or submission readiness.

Sources and artifact trail

Cross-project study (2024) | Authors' artifact
Static prediction study (2022)
CANNIER (2023)

Machine-readable evidence: reports/development-v0.2. Original corpus: dataset/v0.1. Model payloads and transformed source documents are retained locally under ignored local-evidence. See GUIDE_BRIEF.md for meeting talking points.